03. Advanced Sliding Mode Control

Chapter Objectives

This chapter introduce two kinds of advanced sliding mode controllers design

- sliding mode control based on LMI technology
- sliding mode control based on back-stepping technology

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

1. System Description

The kinetic equation of the inverted pendulum

$$\begin{cases} \ddot{\theta} = \dfrac{m(M+m)gl}{(M+m)I+Ml^2}\theta - \dfrac{ml}{(M+m)I+Ml^2}u \\ \ddot{x} = -\dfrac{m^2gl^2}{(M+m)I+Ml^2}\theta + \dfrac{I+ml^2}{(M+m)I+Ml^2}u \end{cases} \quad (3.1)$$

θ : pendulum swing angle θ x : vehicle position

I : moment of inertia, $I = mL^2/12$

g : gravity acceleration, $g = 9.8m/s^2$

M : vehicle mass m : pendulum mass

$2l$: pendulum length u : the control input

Control object: rolling angle $\theta \to 0$, rolling rate $\dot{\theta} \to 0$, the position of vehicle $x \to 0$ and the velocity of vehicle $\dot{x} \to 0$

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

1. System Description

Let $x(1) = \theta$, $x(2) = \dot{\theta}$, $x(3) = x$, $x(4) = \dot{x}$, (3.1) can be written as

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}u \quad (3.2)$$

where, $\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ t_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ t_2 & 0 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ t_3 \\ 0 \\ t_4 \end{bmatrix}$, $D = (M+m)I + Ml^2$

$$t_1 = \frac{m(M+m)gl}{D}, t_2 = -\frac{m^2gl^2}{D}, t_3 = -\frac{ml}{D}, t_4 = \frac{I+ml^2}{D}$$

Considering the uncertainties and the disturbances, (3.2) can be written as

$$\dot{\boldsymbol{x}} = \boldsymbol{A}x + \boldsymbol{B}(u + f(t)) \quad (3.3)$$

where, $\boldsymbol{x} = [\theta \ \dot{\theta} \ x \ \dot{x}]^T$, $|f(t)| \le \delta_f$, $\varepsilon_0 > 0$

$\ddot{\theta} = \frac{m(M+m)gl}{(M+m)I+Ml^2}\theta - \frac{ml}{(M+m)I+Ml^2}u$, $\ddot{x} = -\frac{m^2gl^2}{(M+m)I+Ml^2}\theta + \frac{I+ml^2}{(M+m)I+Ml^2}u$ (3.1)

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

2. Equivalent Sliding Mode Control

The sliding variable is selected as

$$s = \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{x} \quad (3.4)$$

where, $\boldsymbol{P}$ is 4×4 positive-definite matrix

Controller can be designed as

$$u(t) = u_{eq} + u_n \quad (3.5)$$

where, $u_{eq} = -\left(\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right)^{-1}\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{A}x(t)$

$$u_n = -\left(\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right)^{-1}\left(\left|\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right|\delta_f + \varepsilon_0\right)\mathrm{sgn}(s)$$

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

2. Equivalent Sliding Mode Control

Proof

Select the Lyapunov function as $V = s^2/2$

$$\begin{aligned} \dot{s} &= \boldsymbol{B}^T\boldsymbol{P}\dot{\boldsymbol{x}} \\ &= \boldsymbol{B}^T\boldsymbol{P}(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(u+f)) \\ &= \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}u + \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}f \\ &= \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}[-\left(\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right)^{-1}\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{A}\boldsymbol{x} - \left(\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right)^{-1}\left(\left|\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right|\delta_f + \varepsilon_0\right)\mathrm{sgn}(s)] + \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}f \\ &= \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{A}\boldsymbol{x} + [-\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{A}\boldsymbol{x} - \left(\left|\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right|\delta_f + \varepsilon_0\right)\mathrm{sgn}(s)] + \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}f \\ &= -\left(\left|\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right|\delta_f + \varepsilon_0\right)\mathrm{sgn}(s) + \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}f \end{aligned}$$

Therefore

$$\begin{aligned} \dot{V} &= s\dot{s} \\ &= -s\left(\left|\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right|\delta_f + \varepsilon_0\right)\mathrm{sgn}(s) + s\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}f \\ &= -\left|\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right|\delta_f|s| + \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}fs - \varepsilon_0|s| \\ &\le -\varepsilon_0|s| \end{aligned}$$

$s = \boldsymbol{B}^T\boldsymbol{P}\boldsymbol{x}$ (3.4)

$u(t) = u_{eq} + u_n$, $u_{eq} = -\left(\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right)^{-1}\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{A}x(t)$, $u_n = -\left(\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right)^{-1}\left(\left|\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}\right|\delta_f + \varepsilon_0\right)\mathrm{sgn}(s)$ (3.5)

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

3. Sliding Mode Control Based on Auxiliary Feedback

In order to solve for symmetric positive-definite matrix $\boldsymbol{P}$, the controller is designed as

$$u(t) = -\boldsymbol{K}\boldsymbol{x} + v(t) \tag{3.7}$$

where, $v(t) = \boldsymbol{K}\boldsymbol{x} + u_{eq} + u_n$

There exists K such that $\bar{\boldsymbol{A}} = \boldsymbol{A} - \boldsymbol{B}\boldsymbol{K}$ is stable, therefore

$$\dot{\boldsymbol{x}} = \bar{\boldsymbol{A}}\boldsymbol{x} + \boldsymbol{B}(v + f) \tag{3.8}$$

where, $\boldsymbol{K}$ is 1×4 vector

$\boldsymbol{P}$, $\boldsymbol{A}$ and $\bar{\boldsymbol{A}}$ are 4×4 matrixes

$\boldsymbol{B}$ is 4×1 vector

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(u + f(t)) \tag{3.3}$$
$$\boldsymbol{x} = [\theta \;\; \dot{\theta} \;\; x \;\; \dot{x}]^T, |f(t)| \le \delta_f, \varepsilon_0 > 0$$

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

3. Sliding Mode Control Based on Auxiliary Feedback

Proof

Select the Lyapunov function as $V = \boldsymbol{x}^T\boldsymbol{P}\boldsymbol{x}$ (3.9)

Therefore $\dot{V} = \dot{\boldsymbol{x}}^T\boldsymbol{P}\boldsymbol{x} + \boldsymbol{x}^T\boldsymbol{P}\dot{\boldsymbol{x}}$

$= 2\boldsymbol{x}^T\boldsymbol{P}\dot{\boldsymbol{x}} = 2\boldsymbol{x}^T\boldsymbol{P}[\bar{\boldsymbol{A}}\boldsymbol{x} + \boldsymbol{B}(v + f)] = 2\boldsymbol{x}^T\boldsymbol{P}\bar{\boldsymbol{A}}\boldsymbol{x} + 2\boldsymbol{x}^T\boldsymbol{P}\boldsymbol{B}(v + f)$

When $t \ge t_0$, there exits $s = \boldsymbol{B}^T\boldsymbol{P}x = 0$, i.e. $s^T = \boldsymbol{x}^T\boldsymbol{P}\boldsymbol{B} = 0$, such that

$\dot{V} = 2\boldsymbol{x}^T\boldsymbol{P}\bar{\boldsymbol{A}}\boldsymbol{x} = \boldsymbol{x}^T(\boldsymbol{P}\bar{\boldsymbol{A}} + \bar{\boldsymbol{A}}^T\boldsymbol{P})\boldsymbol{x} = 2\boldsymbol{x}^T\boldsymbol{M}\boldsymbol{x}$

In order to satisfy $\dot{V} < 0$, $\boldsymbol{M} < 0$ is needed, i.e.

$\boldsymbol{P}\bar{\boldsymbol{A}} + \bar{\boldsymbol{A}}^T\boldsymbol{P} < 0$

Since $\bar{\boldsymbol{A}}$ is Hurwitz, then $\boldsymbol{P}\bar{\boldsymbol{A}} + \bar{\boldsymbol{A}}^T\boldsymbol{P} < 0$ can be guaranteed

Multiplying $\boldsymbol{P}^{-1}$ in the above inequality

$\bar{\boldsymbol{A}}\boldsymbol{P}^{-1} + \boldsymbol{P}^{-1}\bar{\boldsymbol{A}}^T < 0$

Let $\boldsymbol{X} = \boldsymbol{P}^{-1}$ $\bar{\boldsymbol{A}}\boldsymbol{X} + \boldsymbol{X}\bar{\boldsymbol{A}}^T < 0$ or $(\boldsymbol{A} - \boldsymbol{B}\boldsymbol{K})\boldsymbol{X} + \boldsymbol{X}(\boldsymbol{A} - \boldsymbol{B}\boldsymbol{K})^T < 0$

Select $\boldsymbol{L} = \boldsymbol{K}\boldsymbol{X}$ $\boldsymbol{A}\boldsymbol{X} - \boldsymbol{B}\boldsymbol{L} + \boldsymbol{X}\boldsymbol{A}^T - \boldsymbol{L}^T\boldsymbol{B}^T < 0$ or $\boldsymbol{A}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{A}^T < \boldsymbol{B}\boldsymbol{L} + \boldsymbol{L}^T\boldsymbol{B}^T$ (3.10)

In LMI, to guarantted $\boldsymbol{P}$ as symmetrical matrix, design

$\boldsymbol{P} = \boldsymbol{P}^T$ or $\boldsymbol{X} = \boldsymbol{X}^T$ (3.11)

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

4. Simulation Example (Ex_03.01.m)

The kinetic equation of the inverted pendulum

$$\begin{cases} \ddot{\theta} = \dfrac{m(M+m)gl}{(M+m)I + Mml^2}\theta - \dfrac{ml}{(M+m)I + Mml^2}u \\ \ddot{x} = -\dfrac{m^2gl^2}{(M+m)I + Mml^2}\theta + \dfrac{I + ml^2}{(M+m)I + Mml^2}u \end{cases}$$

Parameters: $g = 9.81m/s^2$, $M = 1kg$, $m = 0.1kg$, $L = 0.5m$

Initial states: $\theta(0) = -\pi/6$, $\dot{\theta}(0) = 0$, $x(0) = 5$, $\dot{x}(0) = 0$

Desired states: $\theta_d = 0$, $\dot{\theta}_d = 0$, $x_d = 0$, $\dot{x}_d = 0$

$f(t) = 0.3\sin t$, $\delta_f = 0.30$

SM Controller: $\varepsilon_0 = 0.15$

Saturated function: $\Delta = 0.05$

Sampling time: $dt = 20ms$

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

4. Simulation Example

The system

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(u + f(t))$$

where, $\boldsymbol{x} = [\theta \;\; \dot{\theta} \;\; x \;\; \dot{x}]^T$, $|f(t)| \le \delta_f$, $\varepsilon_0 > 0$

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ t_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ t_2 & 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ t_3 \\ 0 \\ t_4 \end{bmatrix}, D = (M+m)I + Mml^2$$

$$t_1 = \frac{m(M+m)gl}{D}, t_2 = -\frac{m^2gl^2}{D}, t_3 = -\frac{ml}{D}, t_4 = \frac{I + ml^2}{D}$$

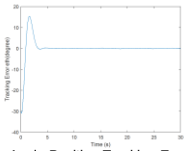
$u(t) = u_{eq} + u_n$

$u_{eq} = -(\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B})^{-1}\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{A}x(t)$

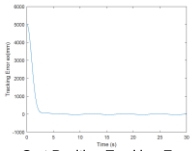
$u_n = -(\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B})^{-1}(|\boldsymbol{B}^T\boldsymbol{P}\boldsymbol{B}|\delta_f + \varepsilon_0)\mathrm{sgn}(s)$

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

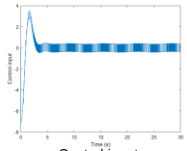
Results with using sign function



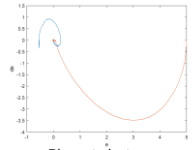
Angle Position Tracking Error



Cart Position Tracking Error



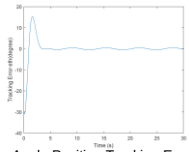
Control input



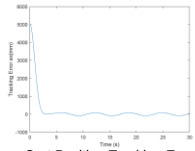
Phase trajectory

§1. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

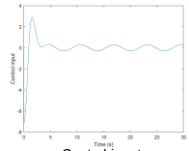
Results with using saturation function



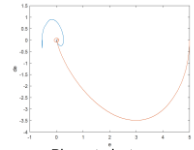
Angle Position Tracking Error



Cart Position Tracking Error



Control input



Phase trajectory

§2. Backstepping Sliding Mode Control for a Inverted Pendulum

1. The Basic Theory

The basic idea of backstepping design is that a complex nonlinear system is decomposed into the subsystems, and the degree of each subsystem doesn't exceed that of the whole system

Accordingly, the Lyapunov function and medial-fictitious control are designed respectively, and the whole system is obtained through "backstepping". Thus the control rule is designed thoroughly

The backstepping method is called as back-deduce method, and the desired dynamic indexes are satisfied

§2. Backstepping Sliding Mode Control for a Inverted Pendulum

2. System Description

The backstepping method and the sliding mode control are integrated to designed a backstepping-sliding-mode controller which realizes the robust control for uncertain systems

Suppose the plant is a nonlinear system as under

$$\begin{cases}\dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x},t) + b(\boldsymbol{x},t)u + d(\boldsymbol{x},t)\end{cases} \quad (3.12)$$

where, $f(\boldsymbol{x},t)$ and $b(\boldsymbol{x},t)$ are the nonlinear functions

$d(\boldsymbol{x},t)$ is the sum of the unmatched uncertainties and the disturbances, and $|d(\boldsymbol{x},t)| \le D$, $b(\boldsymbol{x},t) \ne 0$

§2. Backstepping Sliding Mode Control for a Inverted Pendulum

3. Controller Design

Step 1

Let $e_1 = x_1 - r$ (3.13)

where, r is the desired trajectory. Therefore

$$\dot{e}_1 = \dot{x}_1 - \dot{r} = x_2 - \dot{r}$$

Select the Lyapunov function as

$$V_1 = \frac{1}{2}e_1^2 \quad (3.14)$$

$$\Rightarrow \dot{V}_1 = e_1\dot{e}_1 = e_1(x_2 - \dot{r})$$

In order to realize $\dot{V}_1 \le 0$, let $x_2 = s - c_1e_1 + \dot{r}$, i.e.

$$s = x_2 + c_1e_1 - \dot{r} = c_1e_1 + \dot{e}_1, \quad c_1 > 0 \quad (3.15)$$

where, s is the sliding variable. Therefore

$$\dot{V}_1 = e_1s - c_1e_1^2$$

If $s = 0$, then $\dot{V}_1 \le 0$. Therefore, the next step is required

§2. Backstepping Sliding Mode Control for a Inverted Pendulum

3. Controller Design

Step 2

Select the Lyapunov be as $V_2 = V_1 + \frac{1}{2}s^2$ (3.13)

Because $\dot{s} = \dot{x}_2 + c_1\dot{e}_2 - \ddot{r} = f(x,t) + b(x,t)u + d(t) + c_1\dot{e}_1 - \ddot{r}$

Select the Lyapunov function as

$$\begin{aligned}\dot{V}_2 &= \dot{V}_1 + s\dot{s} \\ &= e_1s - c_1e_1^2 + s(f(x,t) + b(x,t)u + d(t) + c_1\dot{e}_1 - \ddot{r})\end{aligned}$$

In order to realize $\dot{V}_2 \le 0$, a controller is designed as

$$u = \frac{1}{b(x,t)}(-f(x,t) - c_2s - e_1 - c_1\dot{e}_1 + \ddot{r} - \eta\text{sgn}(s)) \quad (3.15)$$

where, $c_2 > 0$, $\eta \ge D$. Therefore

$$\dot{V}_2 = -c_1e_1^2 - c_2s^2 + sd(x,t) - \eta|s| \le 0$$

Therefore, $e_1 \to 0$ and $e_2 \to 0$ as $t \to \infty$

§2. Backstepping Sliding Mode Control for a Inverted Pendulum

4. Simulation Example (Ex_03.02.m)

The dynamic equation of inverted pendulum

$$\begin{cases}\dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u\end{cases}$$

$$f(\boldsymbol{x}) = \frac{g\sin x_1 - mlx_2^2\cos x_1\sin x_1/(m_c+m)}{l[4/3 - m\cos^2 x_1/(m_c+m)]}$$

$$g(\boldsymbol{x}) = \frac{\cos x_1/(m_c+m)}{l[4/3 - m\cos^2 x_1/(m_c+m)]}$$

x_1 : the oscillation angle θ x_2 : the oscillation rate $\dot{\theta}$

g : gravity acceleration, $g = 9.81m/s^2$

m_c : vehicle mass, $m_c = 1kg$ m : pendulum mass, $m = 0.1kg$

$2l$: pendulum length, $2l = 1m$ u : the control input

Desired trajectory $x_d = 0.1sin\pi t$; initial state $[-\pi/60 \quad 0]$

§2. Backstepping Sliding Mode Control for a Inverted Pendulum

4. Simulation Example

The dynamic equation of inverted pendulum

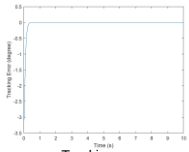
$$\begin{cases}\dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u\end{cases}$$

Desired trajectory $x_d = 0.1sin\pi t$; initial state $[-\pi/60 \quad 0]$

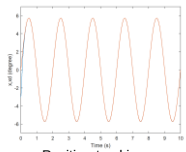
$$e_1 = x_1 - x_d$$
$$c_1 = 35$$
$$c_2 = 15$$
$$\eta = 0.10$$
$$s = c_1e_1 + \dot{e}_1$$
$$u = \frac{1}{g(x,t)}(-f(x,t) - c_2s - e_1 - c_1\dot{e}_1 + \ddot{x}_d - \eta\text{sgn}(s))$$

§2. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

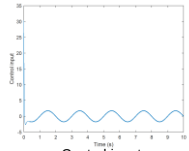
Results with using sign function



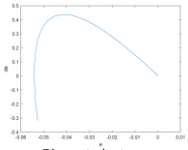
Tracking error



Position tracking



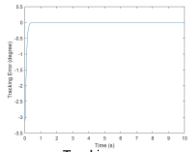
Control input



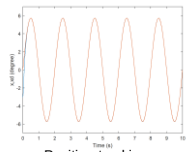
Phase trajectory

§2. SMC Based on a Linear Matrix Inequality for Inverted Pendulum

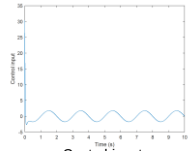
Results with using saturation function



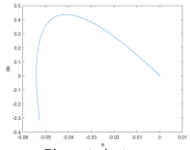
Tracking error



Position tracking



Control input



Phase trajectory